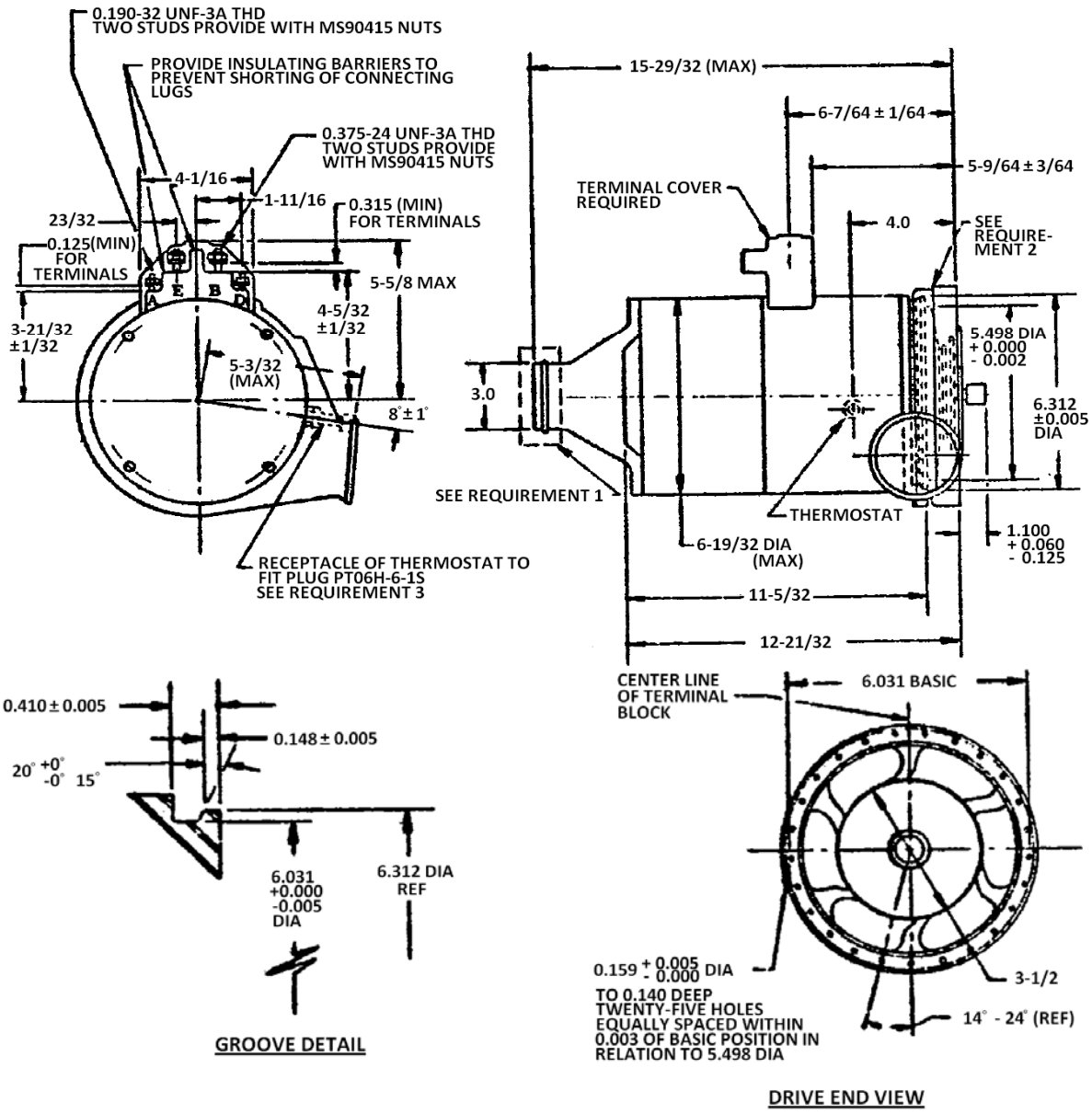


ASSEMBLY	CONSISTS OF		DIRECTION OF ROTATION VIEWED FROM DRIVE END
	GENERATOR	QAD FLANGE & AIR OUTLET	
MS18108-1A	MS18108-1	MS18108-3	CCW
MS18108-2A	MS18108-2	MS18108-3	CW

FOR DESIGN FEATURE PURPOSE, THIS STANDARD TAKE PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.
REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATION FOR BID.

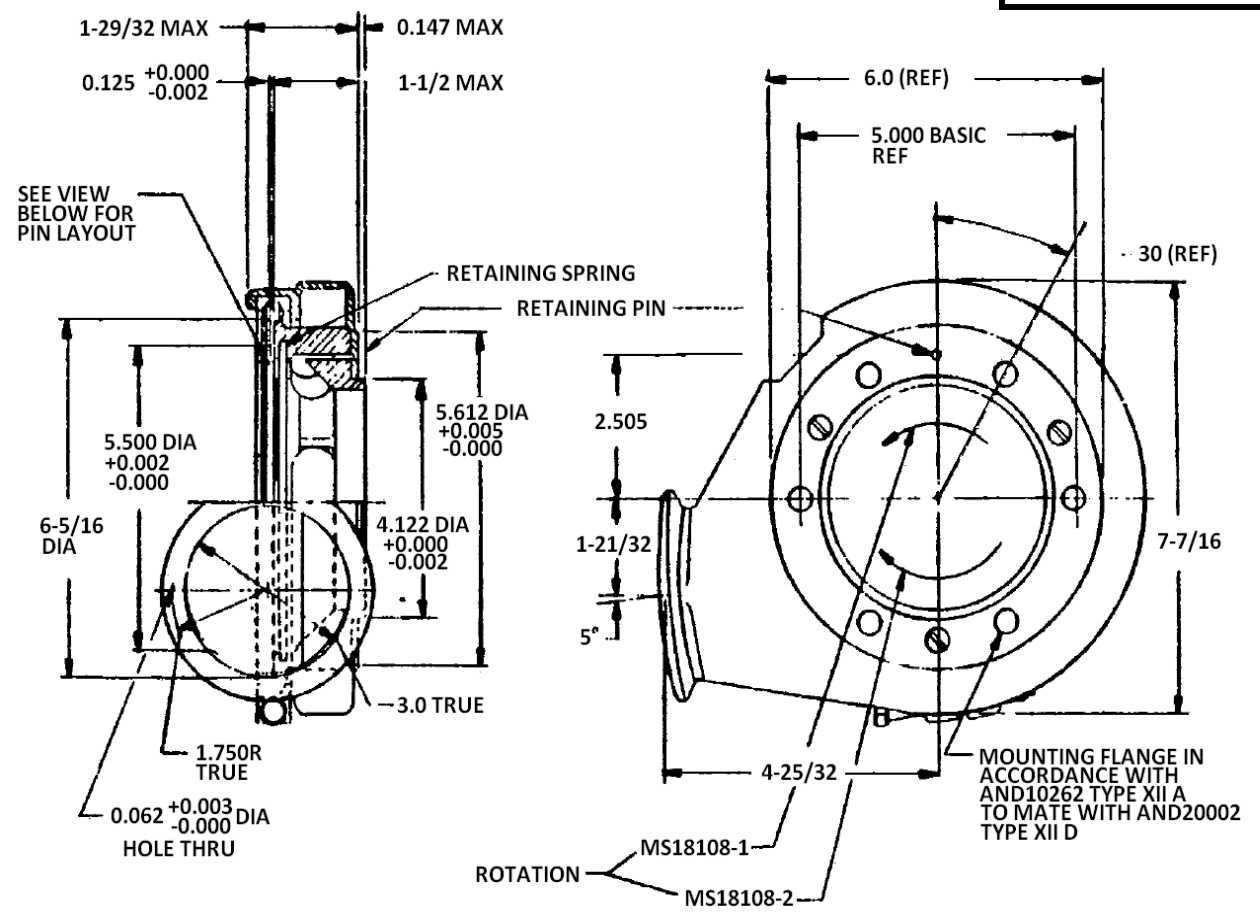
Ⓑ ENTIRE STANDARD REVISED AND REDRAWN

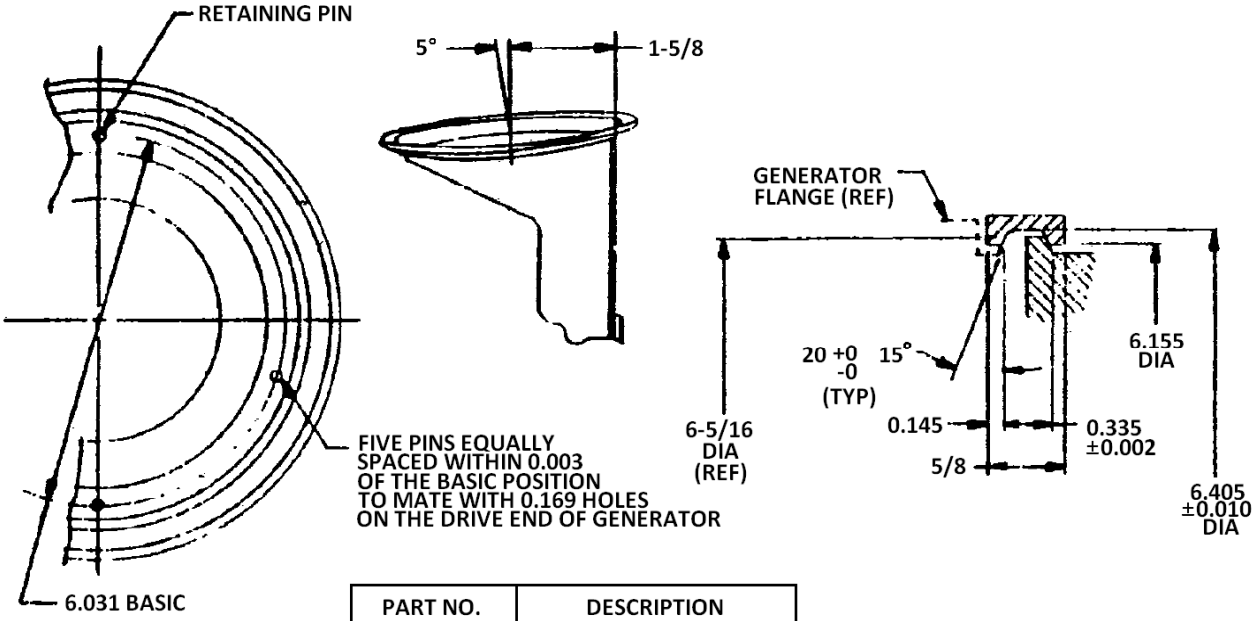
APPROVED 19 OCT 64 REVISED Ⓑ 8 FEB 1974

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P.A. NAVY – AS Other Cust.	TITLE **STARTER-GENERATOR, DIRECT CURRENT, 300 AMPERE, 30 VOLT, WIDE SPEED RANGE, CLASS A, TYPE 3**	**MILITARY STANDARD** **MS18108(AS)**
PROCUREMENT SPECIFICATION MIL-G-6162	SUPERSEDES	SHEET 1 OF 5

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PARTIAL REAR VIEW

PART NO.	DESCRIPTION
MS 18108-3	Q.A.D. FLANGE AND AIR OUTLET

RETAINING RING DETAIL IN LOCKED POSITION

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P.A. NAVY – AS Other Cust.	TITLE STARTER-GENERATOR, DIRECT CURRENT, 300 AMPERE, 30 VOLT, WIDE SPEED RANGE, CLASS A, TYPE 3	MILITARY STANDARD MS18108(AS)
PROCUREMENT SPECIFICATION MIL-G-6162	SUPERSEDES	SHEET 2 OF 5

DD FORM 1 MAR 72 672-1 (Limited Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

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GENERATOR CHARACTERISTICS	
RATED VOLTS	30
RATED CURRENT (NORMAL AMPS)	300
RATED SPEED (RPM)	3000/8000
MINIMUM SPEED FOR REGULATION (RPM)	3500
MAXIMUM SPEED FOR REGULATION (RPM)	10,000
OVERSPEED (RPM)	11,000
REGULATED FIELD CURRENT (MAXIMUM AMPS)	8
POWER DISSIPATED BY REGULATOR (MAX WATTS)	90
WEIGHT (MAXIMUM POUNDS)	56
MOMENT OF OVERHANG (MAXIMUM INCH-POUNDS)	290
MINIMUM NATURAL FREQUENCY (CPS)	250
BLAST COVER (SEE NOTES)	182
SHEAR (POUNDS PER INCH)	1700/2000
EFFICIENCY MINIMUM 70% TO 100% LOAD	70%

REQUIREMENTS:

1. 3” AXIAL SPOUT END PER MS25159
2. Q.A.D. FLANGE AND AIR OUTLET PER SHEET 2
3. THERMOSTAT (SETTING 155 C˚± 3 C˚). USED WITH -1 ONLY.
4. THE AIRCRAFT MANUFACTURER SHALL ALLOW CONNECTION LEAD LENGTH FOR THE LIMITING GENERATOR DIMENSIONS SHOWN ON THIS DRAWING.
5. THE AIRCRAFT MANUFACTURER SHALL ALLOW ADEQUATE CLEARANCE FOR INSTALLING AND REMOVING THE GENERATOR.
6. INTERGRAL FAN REQUIRED FOR SELFCOOLING.
7. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES FRACTIONS 1/32, DECIMALS ± 0.030.
8. GENERATOR SHALL BE FINISHED IN A COLOR CONFORMING TO MILITARY STANDARD 595 COLOR NO. 17875. THE FINISH SHALL BE UNAFFECTED BY ENVIRONMENT CONDITIONS SPECIFIED HEREIN.
9. STARTER PERFORMANCE CRITERIA: STARTING PERFORMANCE CRITERIA IS BASED ON USE OF NAVY 1000 AMPERE, CONSTANT CURRENT, ELECTRICAL POWER SYSTEM DESCRIBED IN MILITARY STANDARD MS17794 (WEP) OF FEBRUARY 1962.
10. STARTER CHARACTERISTICS: THE STARTER-GENERATOR WHILE OPERATING AS A STARTER, WITH 1.25 OHMS EXTERNAL RESISTANCE IN SHUNT FIELD CIRCUIT, SHALL BE CAPABLE OF DELIVERING THE FOLLOWING MINIMUM CRANKING TORQUE AT THE INDICATED DRIVE SPEED (RPM) WHEN POWERED BY THE INDICATED INPUT.

INPUT (AMPERES)	INPUT(VOLTS)	DRIVE-SPEED (RPM)	DRIVE TORQUE (LB-FT)
1000	-	0	82
1000	-	780	82
-	33	1700	44
-	33	2500	0
-	20	1700	0

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11. ENDURANCE: (STARTER) THE STARTER-GENERATOR SHALL BE CAPABLE OF COMPLETING 1200 CYCLES OF OPERATION WHEN OPERATING AS A STARTER IN ACCORDANCE WITH THE FOLLOWING DUTY CYCLE WITH 1.25 OHMS EXTERNAL RESISTANCE IN SHUNT FIELD CIRCUIT. EXTERNAL COOLING MAY BE USED TO EXPEDITE TESTING AND TO PREVENT EXCESSIVE TEMPERATURE. MAXIMUM TIME BETWEEN CYCLES SHALL BE THREE MINUTES.

TABLE I – ENDURANCE TEST – STARTER

FLYWHEEL POLAR MOMENT OF INERTIA LB-FT^2	PHASE A ACCELERATION – FROM REST WITHIN TIME INDICATED			PHASE B STEADY STATE OPERATIONS WITH AUXILIARY TORQUE LOAD AS SPECIFIED				PHASE C ACCELERATION – RPM WHICH FLYWHEEL SHALL BE ACCELERATED WITHIN TIME INDICATED		
	FINAL RPM	INPUT MAX AMPS	MAX. SEC	MIN RPM	DRIVE TORQUE LB-FT	MAX TERM VOLT	TIME SEC	FINAL RPM	TERM VOLT	MAX TIME SEC
91	850	1000	5	850	45±2	24	10	25000	33	15

PHASE A — THIS PHASE SHALL CONSIST OF ACCELERATING THE SPECIFIED FLYWHEEL INERTIA LOAD FROM REST TO THE SPECIFIED SPEED WITHIN THE SPECIFIED TIME AND WITH THE INPUT CURRENT LIMITED TO SPECIFIED VALUE.

PHASE B — THIS PHASE SHALL CONSIST OF A 10 SECOND PERIOD OF STEADY STATE OPERATION AT THE SPECIFIED LOAD AND TERMINAL VOLTAGE.

PHASE C — AFTER COMPLETING PHASE B, THE TORQUE LOAD SHALL BE REMOVED AND THE STARTER PERMITTED TO ACCELERATE THE FLYWHEEL INERTIA LOAD TO THE SPECIFIED SPEED WITHIN THE SPECIFIED TIME.

NOTE — THE ACCELERATIONS SPECIFIED FOR PHASE A AND C ASSUMES A FRICTIONLESS SYSTEM AND DOES NOT CONSIDER TEST STAND LOSSES SUCH AS WINDAGE OR BEARING FRICTION; ACCORDINGLY, IN ORDER TO ENSURE THAT THE MINIMUM DRIVE SPEED SPECIFIED IN PHASE C IS ATTAINED, AN EXTERNAL ASSIST TORQUE MAY BE SUPPLIED OR THE FLYWHEEL INERTIA LOAD MAY BE REDUCED. IN ADDITION, PHASE C OPERATION MAY BE TERMINATED AT A DRIVE SPEED LESS THAN THE SPECIFIED VALUE PROVIDED THE TOTAL HORSEPOWER-SECOND OUTPUT OF THE STARTER IS EQUIVALENT TO THAT WHICH WOULD BE REQUIRED TO ACCELERATE A FRICTIONLESS SYSTEM TO THE SPECIFIED CUT-OFF SPEED.

12. LOW TEMPERATURE: (STARTER) AFTER THE TEMPERATURE OF ALL COMPONENTS OF THE STARTER-GENERATOR HAS BEEN STABILIZED AT -65˚ F (-53.8˚ C) FOR A MINIMUM OF 8 HOURS, THE STARTER-GENERATOR SHALL BE SUBJECTED TO FOUR CONSECUTIVE ENDURANCE TEST CYCLES AS DESCRIBED IN TABLE I. THE PERFORMANCE DEVELOPED DURING PHASE B OF EACH CYCLE SHALL NOT BE LESS THAN 90 PERCENT OF THE VALUES LISTED IN TABLE 1.
13. HIGH TEMPERATURE: (STARTER) REPEAT THE SAME PROCEDURE AS FOR LOW TEMPERATURE EXCEPT THAT THE STABILIZED TEMPERATURE SHALL BE +160˚F (71˚C). SEPARATION OF STARTER AND GENERATOR TEST IS NECESSARY SINCE AN EXTREMELY SOPHISTICATED TEST STAND WOULD BE REQUIRED TO PERMIT STARTING AND GENERATING OPERATION IN QUICK SUCCESSION. PERFORMANCE LOSS SHALL NOT EXCEED THAT ALLOWED IN THE LOW TEMPERATURE TEST.
14. CONSECUTIVE CYCLE: (STARTER) THE STARTER-GENERATOR SHALL BE CAPABLE OF PERFORMING FIVE CYCLES OF OPERATION AT 30 SECONDS INTERVALS. EACH CYCLE SHALL BE IN ACCORDANCE WITH DUTY CYCLE LISTED IN TABLE I. NO EXTERNAL COOLING SHALL BE PROVIDED DURING THE TEST.
15. BRUSH MARKING: EACH BRUSH SHALL HAVE A DIAGONAL GROOVE IN THE ANTI-DRIVE SIDE, EXTENDING FROM THE BRUSH FACE TO A POINT AT THE BRUSH BACK OR FRONT, WHICH INDICATES 100 PERCENT OF ALLOWABLE WEAR.

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16. TERMINALS: THE INTERNAL WIRING OF THE STARTER GENERATOR SHALL TERMINATE IN STAINLESS STEEL TERMINAL STUDS IN A TERMINAL BLOCK PROPERLY MARKED FOR EASY IDENTIFICATION, SUITABLE MS90415 NUTS SHALL BE PROVIDED WITH EACH TERMINAL STUD. INSULATING BARRIERS SHALL BE PROVIDED BETWEEN ADJACENT TERMINALS.
17. PROTECTIVE COVER: A PROTECTIVE COVER OF A FLEXIBLE FIRE RESISTANCE AND AIRCRAFT FLUID RESISTANT MATERIAL SHALL BE PROVIDED FOR THE TERMINAL BLOCK. PROVISIONS SHALL BE MADE TO PERMIT DRESSING WIRES FROM THE TERMINAL BLOCK IN EITHER DIRECTION.
18. OVER-TEMPERATURE: A THERMALLY OPERATED OVER-TEMPERATURE INDICATING SWITCH SHALL BE PROVIDED WITH THE MS18108-1. THE THERMOSTAT SETTING SHALL BE 155˚C ± 3˚C. THE DEVICE SHALL BE PROVIDED WITH A CONNECTOR.
19. BEARINGS: THE ARMATURE OF THE STARTER-GENERATOR SHALL BE SUPPORTED ON BOTH ENDS BY SEALED BEARINGS. THE BEARING SUPPORT ON THE DRIVE SHAFT AND SHALL BE VENTED SO AS TO EQUALIZE PRESSURE ON BOTH SIDES OF THE BEARING. THE STARTER-GENERATOR HOUSING SHALL REMAIN INTACT IN THE EVENT OF PARTIAL OR TOTAL BEARING FAILURE.
20. HEATING: THE ABILITY OF THE GENERATOR TO DELIVER 160 AMPERES, 30 VOLT AT 2960 RPM, 200 AMPERES, 30 VOLTS AT 3050 RPM, AND 250 AMPERES AT 3050 RPM WITH 1.25 OHMS IN EXTERNAL FIELD CIRCUIT AT 40 DEGREE CENTIGRADE AIR IN AND 6.0 INCH H_20 AIR PRESSURE CONTINUOUSLY WITHOUT OVERHEATING SHALL BE DEMONSTRATED.
21. OVERLOAD CAPACITY: THE GENERATOR SHALL BE CAPABLE OF DELIVERING 200 PERCENT OF RATED CAPACITY FOR 5 SECONDS AND 150 PERCENT OF RATED CAPACITY FOR 2 MINUTES AT 6000 RPM.
22. EQUALIZING VOLTAGE: THE EQUALIZING VOLTAGE, UPON TEMPERATURE STABILIZATION, SHALL BE 2.1 ± 0.1 VOLTS MEASURED BETWEEN D AND E TERMINALS.

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